

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Electrical Engineering

2. Subject Name: VLSI Circuits and Systems L-T-P: 3-0-0 Credits:3

3. Pre-requisites:

EE21205 Analog Electronics

EE21208 Digital Electronics

4. Syllabus and reference books:

Syllabus:

Digital Design: Combinational and Sequential designs, Reference generators, Linear regulators, Oscillators, Phase-locked loops (PLLs) and Delay-locked loops (DLLs), Data Converters

Reference Books:

- 1) CMOS VLSI Design: A Circuits and Systems Perspective - by Weste & Harris (4th edition)
- 2) Digital Integrated Circuits by Rabaey, Chandrakasan, Nikolic (2nd edition)
- 3) Analog Integrated Circuit Design, (Second Edition) by Tony Chan Carusone, David Johns, Kenneth Martin, Wiley
- 4) CMOS Analog Circuit Design by Allen and Holberg, Oxford Indian Edition
- 5) Design of Analog CMOS Integrated Circuits by Behzad Razavi, Tata McGraw-Hill
- 6) Power Management Integrated Circuits Architecture, Design and Implementation, Amit Patra, Shailendra Baranwal, Ashis Maity, Samiran Dam, Syed Asif Eqbal Taylor & Francis Ltd, 2024

5. Lecture-wise break-up:

Sl. No.	Topics	No. of lectures
1	Introduction: A few examples of complete systems and identifying the essential modules	2
2	Digital Design: • Review of MOS Transistor basics.	10

	• Delay - RC delay model, logical and electrical effort, delay minimization, method of time constant. • CMOS logic gates - combinational gate design, compound vs. simple gates, layout and fabrication. • Sequential design using CMOS technology - latches, flip flops, timing analysis, metastability, synchronizer. • Power dissipation - static and dynamic power, power reduction.	
3	Reference generators: • Diode/BJT characteristics • Bandgap reference • Current reference, supply independent biasing, temperature independent current reference, voltage to current converter • System level implications.	4
4	Linear regulators: • Low drop-out regulator and frequency compensation • High drop-out regulator and frequency compensation • Package parasitic • System-level considerations	6
5	Oscillators: • Barkhausen Criteria • Ring oscillators • LC oscillators • Voltage controlled oscillators	4
6	Phase-locked loops (PLLs) and Delay-locked loops (DLLs): • Phase detectors • Basic PLL topology • Phase frequency detector • Charge pump • Basic charge pump PLL • Delay-locked loops Basics	6
7	Data Converters: • Sampling theorem	12

	• High-Level specifications (INL, DNL, ENOB, SINAD, THD) • A few examples of ADC realization and Error Correction • A few examples of DAC realization and Error Correction	
	Total Hours	44